Causal Inference
Lecture 1

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based partially on Prof. Arvid Sjölander's slides

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Lecture overview

1. Motivation
   - Nobel prize and chocolate consumption
   - Science and technology and suicides
   - Malta solar power and Masters in Journalism
   - Can astrology be useful?
2. Simpson's paradox
3. Historical notes
4. Statistical associational indices
5. Association
6. Causation
7. Subtle points

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Motivation - does data speaks for itself?

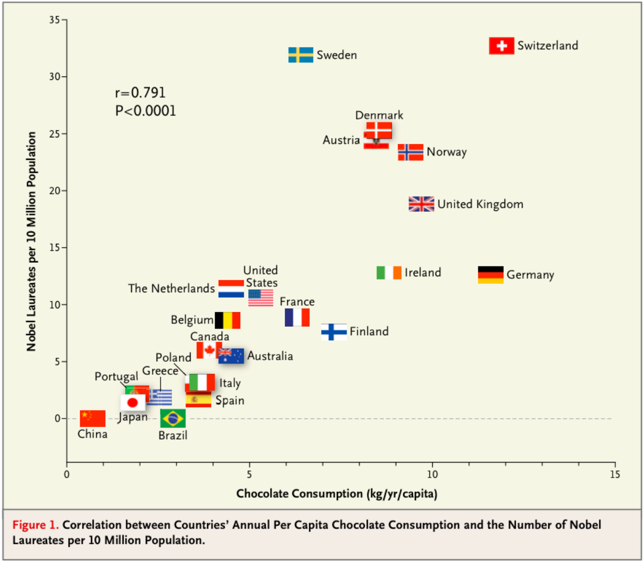

Figure: Chocolate and Nobel Prizes

Eating chocolate will help you winning the Nobel prize?

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Eating chocolate produces Nobel prize winners, says study

By **Oliver Nieburg**
10-Oct-2012 - Last updated on 11-Oct-2012 at 11:51 GMT

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Figure: Chocolate and Nobel Prizes

Geniuses are more likely to eat lots of chocolate?

Forbes

INNOVATION • HEALTHCARE

Chocolate And Nobel Prizes Linked In Study

Larry Husten Former Contributor

I'm a medical journalist covering cardiology news.

Oct 10, 2012, 05:02pm EDT

Updated Nov 17, 2012, 03:29pm EST

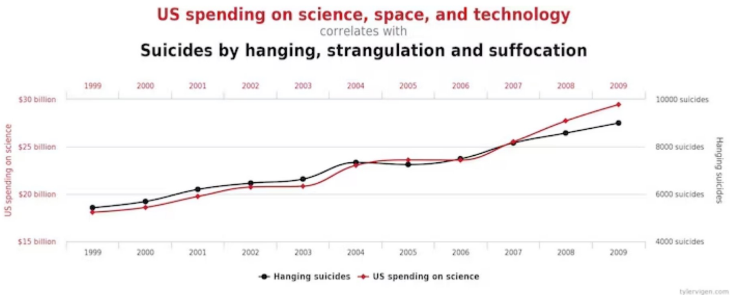
This article is more than 10 years old.

You don't have to be a genius to like chocolate, but geniuses are more likely to eat lots of chocolate, at least according to a new paper published in the August *New England Journal of Medicine*. Franz Messerli reports a highly significant correlation between a nation's per capita chocolate consumption and the rate at which its citizens win Nobel Prizes.

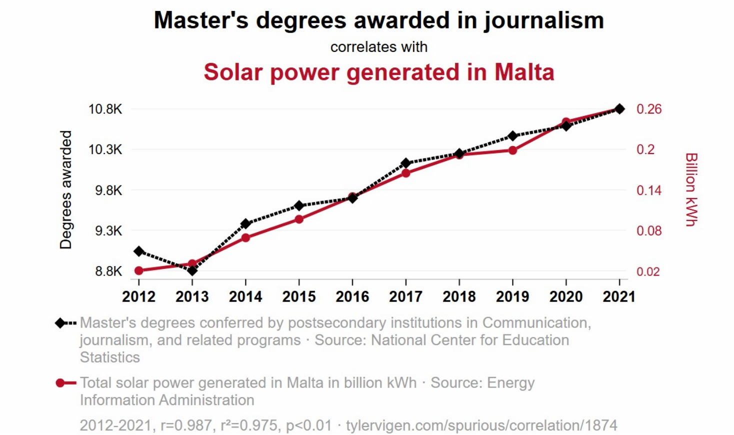
Chocolate sampler (Peter Dazeley/Getty Images)

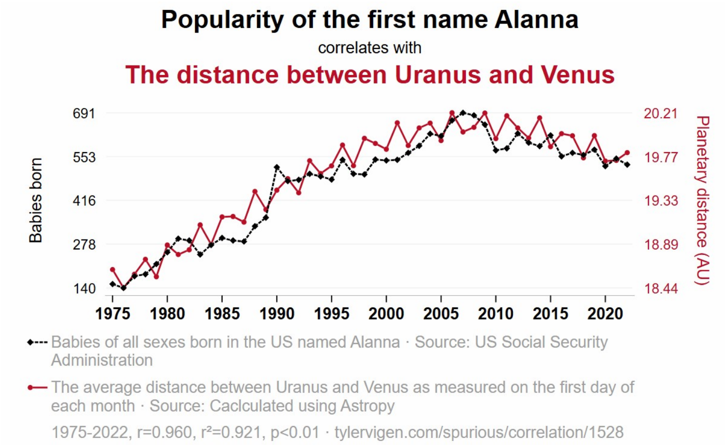
Figure: Chocolate and Nobel Prizes

Spending on science and technology increases suicides by hanging?



Solar power generated in Malta produces journalists?




Popularity of the first name Alanna
correlates with
The distance between Uranus and Venus
Babies born
691
553
416
278
140
Planetary distance (AU)
20.21
19.77
19.33
18.89
18.44
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
Babies of all sexes born in the US named Alanna · Source: US Social Security Administration
The average distance between Uranus and Venus as measured on the first day of each month · Source: Caclculated using Astropy
1975-2022, r=0.960, r²=0.921, p<0.01 · tylervigen.com/spurious/correlation/1528

Current section

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Motivational example

Simpson's paradox

Exposure ($X = x$)	Overall	Men; $G = M$	Women; $G = W$
Treatment A	0.78 (273/350)	0.93 (81/87)	0.73 (192/263)
Treatment B	0.83 (289/350)	0.87 (234/270)	0.69 (55/80)
Risk difference (RD)	-0.05	0.06	0.04

- Treatment A is better for Men and Women.
- When disregarding gender, Treatment B is allegedly better. How it is possible?
- The true marginal RD is

$$\begin{aligned} RD &= RD(M)\mathbb{P}(G = M) + RD(W)\mathbb{P}(G = W) \\ &= 0.06 \times \frac{357}{700} + 0.04 \times \frac{343}{700} \approx 0.052. \end{aligned}$$

The aim in most epidemiological and medical studies

- Estimate the causal effect of the exposure on the outcome.

Problems to overcome

1. The optimal setting for causal inference is the conduction of randomised control trials (RCT). However, most of the epidemiological data is observational.
2. Possible unmeasured confounding variables may jeopardise the estimators' validity (inconsistency).
3. The traditional estimation method deals efficiently only with linear models, whereas most common models in epidemiological, medical and econometrical research are non-linear.

References

1. Hernán, M. A., & Robins, J. M. (2010). Causal inference. Chapter 16.

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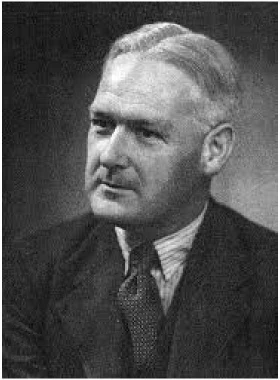
The Bradford Hill criteria for causation (1965)

- Strength of association
- Consistency
- Specificity
- Temporality
- Dose-response relationship
- Plausibility
- Coherence
- Experimental evidence
- Analogy



- The Bradford Hill criteria is a list of (presumably) common features of causal effects
- The criteria do not **define** a causal effect
    - A causal effect may satisfy all or none of the criteria (apart from temporality, which is necessary)
- So what is the definition of a 'causal effect'?

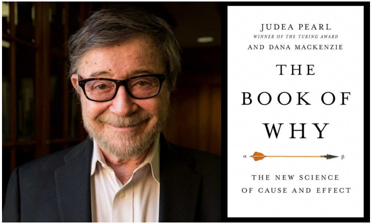
Brief history of causal inference, 70's

- Donald Rubin developed a formal definition of causal effects
  - **potential outcomes**
  - **counterfactuals**
  - `https://www.youtube.com/watch?v=0lpY0Kt4bn8`

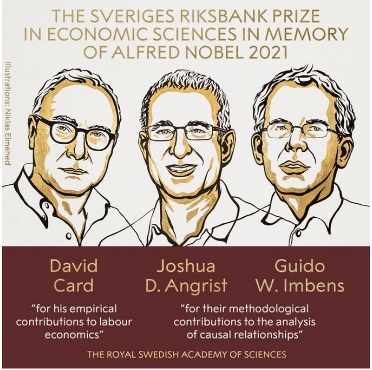
Brief history of causal inference, 80's

- James Robins developed statistical methods for estimating causal effects in longitudinal studies
  - **Marginal Structural Models (MSMs)**
  - **Structural Nested Models (SNMs)**

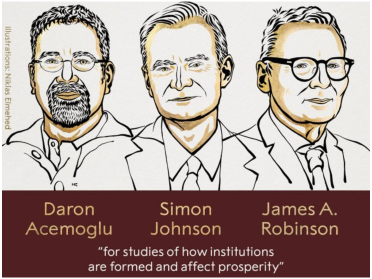
- Judea Pearl developed **Directed Acyclic Graphs (DAGs)**
  - Simplify interpretation and communication in causal inference
  - Useful for covariate selection in observational studies
  - Turing prize of 2011 For fundamental contributions to artificial intelligence through the development of a calculus for probabilistic and causal reasoning.



- Angrist and Imbens - **Instrumental Variables regression**
  - Development of the Local Average Treatment Effect (LATE) Framework
  - Empirical Applications of Econometric Theory



- Acemoglu, Johnson and Robinson -
  - Used natural experiments (colonial histories) and instrumental variables, they provide robust evidence that institutions have a stronger causal effect on economic development than geographic or cultural factors.
  - Use natural experiments (e.g., colonial histories, geography, and historical shocks) to establish the causal link between institutions and economic outcomes.



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Preliminaries

- Suppose we are interested in the relation between an exposure, X, and an outcome, Y
- We assume that both X and Y are binary
    - $X = 0$ for 'unexposed', $X = 1$ for 'exposed, $Y = 0$ for 'does not have the outcome', $Y = 1$ for 'has the outcome'
- We assume that population data are available (infinite sample size)
    - no need for p-values, confidence intervals etc
- These assumptions are pedagogically useful, but often unrealistic
    - will be relaxed later

Joint probability

- Suppose that the population proportions of X and Y are given by

	$Y=0$	$Y=1$
$X=0$	0.88	0.02
$X=1$	0.09	0.01

- Among all subjects, 1% are both exposed and have the outcome
- We say that the **joint probability** of $(X=1, Y=1)$ is 0.01
- We denote this as $p(X=1, Y=1) = 0.01$

Marginal probability

	$Y = 0$	$Y = 1$
$X = 0$	0.88	0.02
$X = 1$	0.09	0.01
$\sum$	0.97	0.03

- Among all subjects, 3% have the outcome
- We say that the **marginal probability** of $Y = 1$ is 0.03
- We denote this as $p(Y = 1) = 0.03$

Conditional probability

	$Y = 0$	$Y = 1$
$X = 0$	0.88	0.02
$X = 1$	0.09	0.01

- Among the exposed subjects, $\frac{0.01}{0.01+0.09} = 10\%$ have the outcome
- We say that the **conditional probability** of having the outcome, among the exposed subjects, is 0.1
- We denote this as $p(Y = 1|X = 1) = 0.1$

A note on terminology

- In this course we mainly use the term 'probability'
- Other synonyms are 'risk' (for negative outcomes) and 'chance' (for positive outcomes)
- For instance, $p(Y = 1|X = 1)$ may be referred to as
  - the probability of the outcome, among the exposed
  - the risk of the outcome, among the exposed
  - the chance of the outcome, among the exposed

Independence and association

- We say that X and Y are **independent** if the probability of the outcome is the same for exposed and unexposed:

$$p(Y=1|X=0) = p(Y=1|X=1) = p(Y=1)$$

- In dense notation:

$$Y \perp\!\!\!\perp X$$

- We say that X and Y are **associated** if the probability of the outcome is different for exposed and unexposed:

$$p(Y=1|X=0) \neq p(Y=1|X=1) \neq p(Y=1)$$

- In dense notation:

$$Y \not\perp\!\!\!\perp X$$

Example

	$Y = 0$	$Y = 1$
$X = 0$	0.88	0.02
$X = 1$	0.09	0.01

- *Are X and Y independent or associated in the table?*

Solution

	$Y=0$	$Y=1$
$X=0$	0.88	0.02
$X=1$	0.09	0.01

$$
\begin{aligned}
p(Y=1|X=0) &= \frac{0.02}{0.02+0.88} = 0.022 \\
p(Y=1|X=1) &= \frac{0.01}{0.01+0.09} = 0.1 \\
p(Y=1) &= 0.02+0.01 = 0.03
\end{aligned}
$$

- $p(Y=1|X=1) \neq p(Y=1|X=0) \neq p(Y=1)$, so X and Y are associated

- The risk difference

$$p(Y=1|X=1) - p(Y=1|X=0)$$

  $Y \amalg X \Leftrightarrow$ risk difference $= 0$

- The risk ratio

$$\frac{p(Y=1|X=1)}{p(Y=1|X=0)}$$

  $Y \amalg X \Leftrightarrow$ risk ratio $= 1$

- The odds ratio

$$\frac{p(Y=1|X=1)}{p(Y=0|X=1)} \bigg/ \frac{p(Y=1|X=0)}{p(Y=0|X=0)}$$

  $Y \amalg X \Leftrightarrow$ odds ratio $= 1$

Example

	$Y = 0$	$Y = 1$
$X = 0$	0.88	0.02
$X = 1$	0.09	0.01

- *Compute the risk difference, the risk ratio, and the odds ratio*

Solution

	$Y=0$	$Y=1$
$X=0$	0.88	0.02
$X=1$	0.09	0.01

$$
\begin{aligned}
p(Y=1|X=0) &= \frac{0.02}{0.02+0.88} = 0.022 \\
p(Y=1|X=1) &= \frac{0.01}{0.01+0.09} = 0.1
\end{aligned}
$$

$$
\begin{aligned}
\text{risk difference} &= p(Y=1|X=1) - p(Y=1|X=0) = 0.078 \\
\text{risk ratio} &= \frac{p(Y=1|X=1)}{p(Y=1|X=0)} = 4.5 \\
\text{odds ratio} &= \frac{p(Y=1|X=1)}{p(Y=0|X=1)} \Big/ \frac{p(Y=1|X=0)}{p(Y=0|X=0)} = 4.89
\end{aligned}
$$

- Sometimes we want to compute the statistical association in separate subgroups of the population - **stratification**
- Let Z be a covariate that defines the strata, e.g Z = sex (0=male, 1=female)
- $p(Y = 1|X, Z)$ is the conditional probability of the outcome, for those with a given level of exposure X and covariate Z
  - e.g. $p(Y = 1|X = 1, Z = 1)$ is the conditional probability of the outcome, among the exposed women, and
  - $p(Y = 1|X = 0, Z = 1)$ is the conditional probability of the outcome, among the unexposed women

Conditional independence and association

- We say that X and Y are conditionally independent, given Z, if

$$p(Y = 1|X = 0, Z) = p(Y = 1|X = 1, Z) = p(Y = 1|Z)$$

  for all values of Z

- In dense notation:

$$Y \perp\!\!\!\perp X \mid Z$$

- We say that X and Y are conditionally associated, given Z, if

$$p(Y = 1|X = 0, Z) \neq p(Y = 1|X = 1, Z) \neq p(Y = 1|Z)$$

  for at least one value of Z

- In dense notation:

$$Y \not\!\perp\!\!\!\perp X \mid Z$$

Measures of conditional association

- Conditional risk difference, given Z

$$p(Y = 1|X = 1, Z) - p(Y = 1|X = 0, Z)$$

- Conditional risk ratio, given Z

$$\frac{p(Y = 1|X = 1, Z)}{p(Y = 1|X = 0, Z)}$$

- Conditional odds ratio, given Z

$$\frac{p(Y = 1|X = 1, Z)}{p(Y = 0|X = 1, Z)} \Big/ \frac{p(Y = 1|X = 0, Z)}{p(Y = 0|X = 0, Z)}$$

- All common causal models are essentially equivalent, from a mathematical perspective
    - different languages, same content
- To define 'causation', we will mostly rely on the potential outcome model, but borrow from the other models as well

Motivating example

- August has smoked 5 cigs/day since he was 15 years old. At the age of 60 he develops lung cancer
- *Did the smoking cause the cancer?*

Important information to consider

- Unhealthy work environment?
- Genetic predisposition?
- Unhealthy diet?
- Alcohol habits?
- etc etc
- *What more information do we need?*

- What would have happened to August, had he not smoked, everything else equal
- If he would not then have gotten cancer, then the smoking must have contributed

Human reasoning about cause and effects

- We mentally compare the outcome under two scenarios:
  - the exposure is present
  - the exposure is absent

  everything else equal
- If the outcome differs under these scenarios, then we say that the exposure has a causal effect
  - causative or preventative

- Let Y_x be the outcome that we would observe, for a given subject, if the subject potentially received exposure level x
    - Y_0 is the outcome when not exposed
    - Y_1 is the outcome when exposed
- Y_0 and Y_1 are referred to as **potential outcomes**
- Ideally - **and very unrealistically** - we could observe both potential outcomes for any given subject

subject	Y_0	Y_1
August	0	1
Selma	0	0
Fjodor	1	1

Subject-specific causal effects

subject	Y_0	Y_1
August	0	1
Selma	0	0
Fjodor	1	1

- X has a causal effect on Y, for a given subject, if the potential outcomes Y_0 and Y_1 differ for this subject
  - for August, the exposure has an effect: $Y_0 \neq Y_1$
  - for Selma and Fjodor, the exposure has no effect; $Y_0 = Y_1$

Observed data

- August is exposed ($X = 1$). Thus, for August
  - Y_1 is observed and equal to the factual outcome Y
  - Y_0 is unobserved, or **counterfactual**
- Selma and Fjodor are unexposed ($X = 0$). Thus, for Selma and Fjodor
  - Y_0 is observed and equal to the factual outcome Y
  - Y_1 is unobserved, or **counterfactual**

subject	X	Y	Y_0	Y_1
August	1	1	?	1
Selma	0	0	0	?
Fjodor	0	1	1	?

A fundamental problem of causation

- It is very difficult to say whether the exposure has a causal effect on the outcome for a specific subject
  - because we cannot observe the same subject under two exposure levels simultaneously

From subjects to populations

- Fortunately, it is much easier to justify causal claims on population levels
  - e.g. ‘if nobody would smoke, then the incidence of lung cancer would be 15% less than if everybody would smoke’
  - more later

Population causal effects

- $p(Y_x = 1)$ is the probability of the outcome if **everybody** would receive exposure level x
- X has a population causal effect on Y if

$$p(Y_0 = 1) \neq p(Y_1 = 1)$$

- X has no population causal effect on Y if

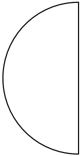
$$p(Y_0 = 1) = p(Y_1 = 1)$$

- Hereafter we refer to 'population causal effects' as 'causal effects'

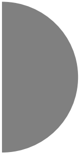
Association vs Causation

- Association:

Factually unexposed



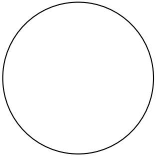
Factually exposed



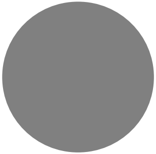
$$p(Y = 1|X = 0) \text{ vs } p(Y = 1|X = 1)$$

- Causation:

Everybody unexposed



Everybody exposed



$$p(Y_0 = 1) \text{ vs } p(Y_1 = 1)$$

Measures of causal effect

- The causal risk difference

$$p(Y_1 = 1) - p(Y_0 = 1)$$

  no causal effect of X on Y $\Leftrightarrow$ causal risk difference $= 0$

- The causal risk ratio

$$\frac{p(Y_1 = 1)}{p(Y_0 = 1)}$$

  no causal effect of X on Y $\Leftrightarrow$ causal risk ratio $= 1$

- The causal odds ratio

$$\frac{p(Y_1 = 1)}{p(Y_1 = 0)} \Big/ \frac{p(Y_0 = 1)}{p(Y_0 = 0)}$$

  no causal effect of X on Y $\Leftrightarrow$ causal odds ratio $= 1$

Example

subject	Y_0	Y_1
1	0	0
2	0	1
3	0	0
4	1	1
5	0	0
6	1	1
7	1	1
8	1	1
9	0	0
10	0	1

- *Compute the causal risk difference, the causal risk ratio, and the causal odds ratio*

Solution

subject	Y_0	Y_1
1	0	0
2	0	1
3	0	0
4	1	1
5	0	0
6	1	1
7	1	1
8	1	1
9	0	0
10	0	1

$$\begin{aligned} p(Y_0 = 1) &= 4/10 = 0.4 \\ p(Y_1 = 1) &= 6/10 = 0.6 \end{aligned}$$

$$\begin{aligned} \text{causal risk difference} &= p(Y_1 = 1) - p(Y_0 = 1) \\ &= 0.2 \\ \text{causal risk ratio} &= \frac{p(Y_1 = 1)}{p(Y_0 = 1)} = 1.5 \\ \text{causal odds ratio} &= \frac{p(Y_1 = 1)}{p(Y_1 = 0)} \Big/ \frac{p(Y_0 = 1)}{p(Y_0 = 0)} \\ &= 2.25 \end{aligned}$$

Measures of conditional causal effect

- Conditional causal risk difference, given Z

$$p(Y_1 = 1|Z) - p(Y_0 = 1|Z)$$

- Conditional causal risk ratio, given Z

$$\frac{p(Y_1 = 1|Z)}{p(Y_0 = 1|Z)}$$

- Conditional causal odds ratio, given Z

$$\frac{p(Y_1 = 1|Z)}{p(Y_1 = 0|Z)} \Big/ \frac{p(Y_0 = 1|Z)}{p(Y_0 = 0|Z)}$$

Brief remark #1

- We have seen that both association and causation can be quantified with risk differences, risk ratios, and odds ratios
- For convenience, we will mostly focus on risk ratios
- Everything that we say holds for risk differences and odds ratios as well

Brief remark #2

- We have considered binary variables and defined the (population) causal effect as

$$p(Y_0 = 1) \text{ vs } p(Y_1 = 1)$$

- When the outcome is continuous we may define the causal effect as

$$E(Y_0) \text{ vs } E(Y_1)$$

- When the exposure is continuous as well, we may define the causal effect of an increase from $X = 0$ to $X = x$ as

$$E(Y_0) \text{ vs } E(Y_x)$$

When is a counterfactual well defined?

- Some people have argued that counterfactuals are not always well defined
  - i.e. there is no uniform agreement on what the counterfactual represents 'in real life'
- If these people are correct, then causal research questions based on these counterfactuals are not well defined either

Example

- Causal research question: does obesity have a causal effect on the risk of cardiovascular disease?
- Define $X = 1$ if BMI>30, and $X = 0$ if BMI<30
- Cardiovascular disease (Y) is more common among obese than among non-obese, i.e.
$$p(Y = 1|X = 1) > p(Y = 1|X = 0)$$
- Causal contrast of interest
$$p(Y_0 = 1) \text{ vs } p(Y_1 = 1)?$$

Quite a vague question

- 'everybody exposed' = 'everybody BMI>30'
- But what does 'if everybody had BMI>30' really mean?
  - everybody fat or everybody muscular?
  - fat on the belly or fat on the hips?
- The outcome is probably very different under these alternative counterfactual scenarios
  - thus, the counterfactuals that we wish to compare are not well defined

An important difference between association and causation

- The association between X and Y is well defined if we agree on
  - who is factually exposed (e.g. BMI>30), and
  - who does factually have the outcome (e.g. cardiovascular disease)
- This is typically uncontroversial
- The causal effect of X on Y is well defined if we also agree on
  - the counterfactual scenario where all subjects are exposed
  - the counterfactual scenario where all subjects are unexposed
- This is often more difficult

Refining the research question

- To be more precise we may refine the research question
  - e.g. 'what is the causal effect on the risk of cardiovascular disease, of having 20% body fat as compared to 10% body fat, when 30% of all body fat is located on the belly and 30% is located on the hips?
- But refining the research question quickly limits
  - the relevance of it, since it becomes 'too narrow' for general interest
  - the possibility to answer it, since very few people would 'match' our inclusion criteria
- A trade-off between preciseness and relevance/feasibility

Summary

- Association is not necessarily equal to causation
- To define causation, we use potential outcomes and counterfactuals
- Beware: not all counterfactuals (and causal effects) are well defined